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1.2 Step-Wise Project Planning — Each Step in 2 Points

Definition: Project planning is the process of thinking through and organizing the activities required to achieve a desired goal.

The Step-Wise project planning framework has Steps 0 to 10.

Step 0: Select Project

Select the project based on its priority and strategic importance.

A feasibility study may show that a project is worthwhile, but it must be compared with other projects.

Step 1: Identify Project Scope and Objectives

Define the project objectives, measures of success, stakeholders, and project authority.

Establish communication methods and modify objectives when necessary based on stakeholder interests.

Step 2: Identify Project Infrastructure

Identify the project's relationship with strategic planning and document required hardware/software standards.

Establish standards for installation, quality, change control, configuration management, and project organization.

Step 3: Analyse Project Characteristics

Determine whether the project is objective-driven or product-driven and identify major project risks.

Select a suitable development methodology and life-cycle approach, then review resource estimates.

Step 4: Identify Project Products and Activities

Identify project products/deliverables and organize them using a Product Breakdown Structure (PBS).

Create product flows and an activity network showing the order of tasks and milestones.

Step 5: Estimate Effort for Each Activity

Estimate staff effort, elapsed time, and non-staff resources for every activity.

Break very long activities into smaller, controllable subtasks for easier monitoring.

Step 6: Identify Activity Risks

Identify and prioritize risks according to their likelihood and possible damage.

Plan risk-reduction and contingency measures and adjust estimates accordingly.

Step 7: Allocate Resources

Identify and allocate the required people and other resources to activities.

Adjust the plan for resource constraints and prepare a Gantt chart showing activity timing and concurrency.

Step 8: Review / Publicize Plan

Review the plan's quality criteria ar ... ate quality checks.

Document the plan and obtain agre ... mmitments.

Step 9: Execute Plan

Start carrying out the activities acc ... n.

Monitor progress and use more detailed planning as activities approach their execution stage.

Step 10: Lower Levels of Planning

Prepare detailed plans for individual activities as they become due.

Update provisional plans using the more accurate information available closer to the activity start.

1.3 Software Quality – TQM and Six Sigma

TQM – Total Quality Management

Definition: TQM is a management philosophy where every member of an organization is responsible for continuously improving quality of products, processes, and services to achieve long-term customer satisfaction.

Main Points

Customer Focus – Quality means satisfying customer requirements and expectations.

Continuous Improvement (Kaizen) – Prevent defects by continuously improving the process.

Total Employee Involvement – Everyone is responsible for quality, not only the QA team.

Fact-Based Decisions – Use measured data rather than opinions for improvement.

Process-Centred Approach – Quality management is integrated into all organizational processes.

Six Sigma

Definition: Six Sigma is a data-driven statistical approach used to reduce process variation and defects. It targets 3.4 defects per million opportunities (DPMO).

DMAIC Cycle

D – Define

Define the problem and project goals.

Identify customer requirements.

M – Measure

Measure the current process performance.

Collect relevant data.

A – Analyze

Analyze the collected data.

Find the root causes of defects or variation.

I – Improve

Develop improvements to the process.

Eliminate or reduce the root causes of problems.

C – Control

Monitor the improved process.

Maintain the improvements using control mechanisms.

ISO 9126 and External Standards — 13 Marks

1. ISO/IEC 9126

ISO/IEC 9126 is an international standard for evaluating software quality. It defines a software quality model with six major quality characteristics.

1. Functionality – Ability of software to provide functions that meet stated and implied needs.

Sub-characteristics: suitability, accuracy, interoperability, security and compliance.

2. Reliability – Ability of software to maintain its performance under specified conditions for a period of time.

Sub-characteristics: maturity, fault tolerance, recoverability and compliance.

3. Usability – Effort required to use the software and users' satisfaction with it.

Sub-characteristics: understandability, learnability, operability, attractiveness and compliance.

4. Efficiency – Relationship between software performance and the resources used.

Sub-characteristics: time behaviour and resource utilization.

5. Maintainability – Ease and effort required to modify the software.

Sub-characteristics: analyzability, changeability, stability, testability and compliance.

6. Portability – Ability of software to be transferred from one environment to another.

Sub-characteristics: adaptability, installability, co-existence, replaceability and compliance.

> Note: ISO/IEC 9126 was later superseded by the ISO/IEC 25000 (SQuaRE) series, but its six-characteristic model is still widely referenced.

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2. External Standards

External standards are quality and process standards defined outside an organization by standards bodies, industry groups, or regulatory authorities.

Important external standards:

ISO 9001

It is a quality management system standard.

It requires documented processes, quality records, and continuous improvement.

CMMI

Capability Maturity Model Integration is a process-improvement framework.

It has 5 maturity levels: Initial, Managed, Defined, Quantitatively Managed, and Optimizing.

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3. ISO/IEC 9126 / ISO/IEC 25000 – Defines software quality characteristics and their evaluation.

4. IEEE Standards – Examples include IEEE 830 for software requirements specifications and IEEE 730 for software quality assurance plans.

5. ISO/IEC 12207 – Defines processes and activities involved in the software life cycle.

Software Quality

Software quality means that software conforms to its functional and performance requirements, development standards, and expected characteristics.

Two important dimensions

Quality of Design – Characteristics specified during requirements, specification, and system design.

Quality of Conformance – How well the implemented software follows the design and meets requirements and performance goals.

1.1 Software Project Management

■ It involves planning, monitoring, and controlling people, processes, and events during software development.

■ Its main aim is to deliver high-quality software on time and within budget. Important Points

People

Involves managers, developers, customers and end-users.

Includes recruitment, training, motivation and team development.

Product

Defines the product objectives and software scope.

Considers functions, performance and alternative solutions.

Process

Defines the framework activities, tasks, milestones and work products.

The selected process model should suit the product and project environment.

Project

Involves planning, monitoring and controlling the project.

Ensures resources, cost, schedule and quality are properly managed.

Software project vs other types of project

invisibility – Software progress is not physically visible.

Complexity – Software products can contain very high complexity.

Conformity – Software must conform to human requirements, which may be inconsistent.

Flexibility – Software can be changed easily and is expected to adapt to changing environments.

Project – Characteristics, Activities Covered & Types of Projects

1. Project – Definition

A project is a planned undertaking in which the work is determined before it starts.

It is carried out to achieve a specific objective or create a specified product within a given time and available resources.

2. Characteristics of a Project

It involves non-routine tasks and requires proper planning.

It has specific objectives/products and a predetermined time span.

It involves limited resources and may require several specialisms.

It is carried out in several phases and may be large or complex.

3. Activities Covered by Software Project Management

Software project management activities follow a Feasibility Study → Planning → Execution cycle.

1. Feasibility Study
Determines whether the proposed software project is worth starting.
Estimates development/operational costs and compares them with expected benefits.
2. Planning
Creates a plan for achieving the project goals.
For large projects, the overall plan is prepared first, while detailed plans are made as each stage approaches.

3. Project Execution
Project execution generally includes the following activities:
Requirements Analysis – Find out in detail what users require.
Specification – Document what the proposed system should do.
Design – Design the external/user interface and internal structure of the system.
Coding – Write the software program.
Verification and Validation – Test whether the system meets its requirements.
Implementation / Installation – Install the developed system.
Maintenance and Support – Correct errors and provide improvements after implementation.

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Managing Contracts –
1. Contract Planning
Defines the scope, requirements, budget, schedule, and responsibilities.
Selects the suitable contract type for the project.
2. Supplier Selection
Identifies and evaluates suppliers based on cost, experience, quality, and capability.
Selects the supplier who best meets the project requirements.
3. Contract Negotiation
Negotiates price, schedule, scope, payment, and responsibilities.
Both parties agree to the terms and sign the contract.
4. Contract Execution
Supplier starts the work according to the agreed contract.
Project manager ensures that work is completed according to scope, quality, and schedule.
5. Contract Monitoring and Control
Monitors cost, time, quality, deliverables, and supplier performance.

Manages changes, risks, and disputes during the contract.

6. Contract Closure

Confirms that all deliverables are completed and accepted.

Completes final payment, documentation, and lessons learned.

Types of Contracts

1. Fixed-Price Contract

A fixed price is agreed for the complete project.

Suitable when requirements are clear and stable.

2. Cost-Reimbursable Contract

Customer pays the actual project costs plus an agreed fee.

Suitable when requirements are uncertain.

3. Time and Materials Contract

Payment is based on time spent and materials used.

Suitable for projects where the scope may change.

4. Unit-Price Contract

A fixed price is given for each unit of work.

Total cost depends on the number of units delivered.

Cost-Benefit Analysis (CBA) – 2 Point Answers

1. Definition of Cost-Benefit Analysis

Cost-Benefit Analysis (CBA) is a systematic method of identifying, estimating and comparing project costs and benefits.

It helps determine the overall value and feasibility of a project.

2. Stages of CBA

Stage 1: Identify and estimate all costs and benefits during development

Stage 2: Convert them into a common unit, usually money, for comparison.

3. Development Costs

These are costs incurred while building the system.

Examples include staff effort, hardware, software licences and other development resources.

4. Setup/Implementation Costs

These are costs involved in installing and introducing the system.

Examples include installation, data conversion, user training and change-over costs.

5. Operational Costs

These are the ongoing costs incurred after the system becomes operational.

Examples include maintenance, technical support and consumables.

6. Direct/Tangible Benefits

These are benefits that can be measured and expressed financially.

Examples include reduced staff costs, fewer errors and faster processing.

7. Indirect/Intangible Benefits

These are benefits that are difficult to measure in monetary terms.

Examples include improved customer satisfaction, better decision-making and enhanced company image.

8. Difficulty in Cost-Benefit Analysis

A major difficulty is that some costs and benefits are intangible or uncertain.

Therefore, reasonable efforts should be made to quantify them approximately and clearly distinguish estimates from firm commitments.

Cost-Benefit Evaluation Techniques – 2 Point Answers

1. Net Profit

Net profit is the difference between total benefits and total costs over the life of a project.

It is simple to calculate but does not consider the timing of cash flows or investment size.

Formula :Net Profit = total benefits - total costs

2. Payback Period

Payback Period is the time required to recover the initial investment from cash inflows.

A shorter payback period is generally preferred.

Formula: Payback Period = Initial Investment / Annual Cash Inflow

3. Return on Investment (ROI) / Accounting Rate of Return

ROI measures the average annual profit as a percentage of the investment.

It allows easy comparison between projects but does not consider the timing of cash flows.

Formula: ROI = (Average Annual Profit ÷ Total Investment) × 100

4. Net Present Value (NPV)

NPV measures the difference between the present value of benefits and present value of costs.

NPV > 0 means the project is generally acceptable.

Formula: NPV = PV of Benefits − PV of Costs

5. Internal Rate of Return (IRR)

IRR is the discount rate at which the NPV of a project becomes zero.

The project is generally accepted when IRR > required rate of return.

Formula: NPV = 0 at IRR

Cash Flow Forecasting

Cash flow forecasting is the process of estimating the costs (cash outflows) and benefits (cash inflows) of a software project for each period, usually year by year, throughout the project’s life.

Main Components

Development/Investment Costs – Money spent to develop and implement the system.

Operational Costs – Ongoing expenses such as maintenance and support.

Benefits – Financial gains obtained after the system becomes operational.

Net Cash Flow – Difference between benefits and costs for a particular period.

Cash Flow Statement

2. Example

Suppose a software project has the following forecast:

If the project starts with ■2,00,000, the closing cash balance would be:

January: ■2,00,000 + ■1,50,000 = ■3,50,000

February: ■3,50,000 – ■50,000 = ■3,00,000

March: ■3,00,000 + ■1,00,000 = ■4,00,000

This shows the project is expected to maintain a positive cash balance.

Definition: Risk evaluation is the process of assessing the uncertainty in project costs, benefits and outcomes and determining how these risks affect the project's financial and strategic attractiveness.

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Part D: Case Study – PRINCE2 – 2 Point Answers

1. Overview of PRINCE2

PRINCE2 (PRojects IN Controlled Environments) is a structured, process-based project management methodology. used for IT and other types of projects.

It provides a scalable framework based on 7 principles, 7 themes and 7 processes.

Seven Principles

2. Continued Business Justification

A project must have a valid business reason for starting and continuing.

reviewed throughout the project.

3. Learn from Experience

Lessons from previous and current projects should be identified, recorded and applied.

It improves project performance.

4. Defined Roles and Responsibilities

PRINCE2 clearly defines roles, responsibilities and accountability.

5. Manage by Stages

The project is divided into manageable stages.

Each stage is planned, monitored and controlled before moving to the next stage.

6. Manage by Exception

Tolerances are established for time, cost, quality, scope, risk and benefits.

7. Focus on Products

PRINCE2 emphasizes defining and delivering required products/deliverables.

8. Tailor to Suit the Project Environment

PRINCE2 should be adapted according to the project's size, complexity, risk and organizational environment.

Seven Themes

9. Business Case

The Business Case explains the reason, costs, benefits and justification for the project.

Ensures the project remains justified.

10. Organization

Defines the project management structure, roles and responsibilities.

11. Quality

Defines the required quality standards, criteria and controls for project products.

Ensures products meet the required quality.

12. Plans

Defines how project plans are developed.

Helps manage activities, time and resources.

13. Risk

Risk management involves identifying, assessing and controlling uncertainty.

14. Change

Identifies and assesses potential changes.

Controls approved changes to project baselines.

15. Progress

Progress involves comparing actual achievements with planned targets.

Significant deviations are reported.

Seven Processes

1. Starting Up a Project (SU)

Checks if the project is worthwhile and feasible.

Ensures resources are not committed unnecessarily.

2. Directing a Project (DP)

Gives the Project Board overall control.

Delegates day-to-day work to the Project Manager.

3. Initiating a Project (IP)

Establishes a strong foundation for the project.

Produces the Project Initiation Documentation (PID).

4. Controlling a Stage (CS)

Covers day-to-day project monitoring and control.

Ensures stage activities stay within plan.

5. Managing Product Delivery (MP)

Controls work between Project Manager and Team Manager.

Ensures products are accepted, created and delivered.

6. Managing a Stage Boundary (SB)

Reviews the current stage's performance.

Plans the next stage and confirms continued justification.

7. Closing a Project (CP)

Confirms project products are accepted.

Ensures objectives are achieved or the project is properly closed.

23. Relevance of PRINCE2

PRINCE2 provides discipline, traceability, clear accountability and staged control, especially useful for large or complex projects.

Since it is process-based rather than tool-specific, it can be implemented using different project management software tools.

Evolution of Globalization & Global Teams – 2 Point Answers

1. Globalization

Globalization is the increasing interconnection of economies, markets, businesses and cultures across countries.

In project management, it enables organizations to staff and execute projects across different countries and time zones.

■ Pre-1990s: Projects were mainly local with teams working in the same office.

■ 1990s: Multinational companies and outsourcing created cross-border teams.

■ Late 1990s–2000s: Internet, email, and broadband enabled global collaboration.

■ 2000s: Outsourcing and offshoring became common to reduce costs and access skills.

■ 2010s–Present: Cloud tools, video conferencing, and virtual teams enable fully distributed projects.Key point: Modern project managers need technical skills as well as cultural, communication, and coordination skills.

2. Challenges in Building Global Teams

Time Zone Differences – Different time zones make it difficult to schedule meetings and obtain quick responses.

Cultural Differences – Differences in language, communication style, hierarchy, decision-making and work practices can cause misunderstandings.

Communication Barriers – Language differences and lack of face-to-face communication can lead to misinterpretation.

Trust Building – It is harder to develop trust when team members rarely meet personally.

Technology and Infrastructure Gaps – Differences in internet connectivity, hardware and collaboration tools can affect productivity and teamwork.

Legal and Regulatory Differences – Different countries have different labour, data protection, intellectual property, and contract laws.

Coordination and Control – Managing progress, quality, and information is harder when teams are spread across locations.

Team Cohesion and Motivation – Remote workers may feel isolated, reducing engagement and productivity.

3. Models for Managing Global Teams

1. Centralized Model

A central team at headquarters controls and directs work done by teams in other countries.

2. Decentralized Model

Regional teams have authority to manage their own part of the project.

3. Hybrid Model

A central team provides overall direction, while regional teams have flexibility to work according to local needs.

4. Virtual Team Model

Team members work mainly through digital tools without being tied to a physical office.

4. Techniques for Managing Global Teams

Clear Communication Protocols – Decide which communication tools to use and set response-time expectations.

Overlapping Working Hours – Create common working hours for meetings and rotate meeting times fairly.

Cultural Awareness Training – Help employees understand and respect different cultures and working styles.

Build Trust – Use regular meetings, informal interaction, and transparent communication to develop trust.

Standardize Processes and Tools – Use common methods, templates, and collaboration platforms.

Clear Roles and Responsibilities – Define who is responsible for each task and how problems are escalated.

Strong Leadership – The project manager should clearly communicate the vision and recognize contributions from all locations.

Monitor and Adapt – Regularly evaluate the team and change tools, communication, or structure when needed.

Part B: Impact of the Internet on Project Management – 2 Point Answers

1. Introduction

The internet has changed how projects are planned, executed, monitored and closed.

It removes geographical and communication barriers and enables new internet-based projects.

2Effects of the Internet on Project Management

Real-Time Communication – Email, chat, and video calls allow instant communication between stakeholders anywhere.

Access to Information – Shared documents, dashboards, and online repositories provide information from any location.

Remote and Virtual Teams – Teams can work together even when members are in different locations or countries.

Online Project Management Tools – Online tools help with scheduling, task tracking, resource management, and reporting.

Faster Stakeholder Engagement – Clients and sponsors can review work, give feedback, and approve deliverables online.

Increased Risk Exposure – Internet use creates risks such as cyberattacks, data breaches, and service outages.

Documentation and Version Control – Cloud storage and collaborative editing help teams use the latest version of documents.

Hackman–Oldham Job Characteristics Model

This model identifies five core job characteristics that influence employee motivation, satisfaction and performance:

• Skill Variety: The degree to which a job requires a range of different skills and talents.

• Task Identity: The degree to which the job involves completing a whole, identifiable piece of work from beginning to end.

• Task Significance: The degree to which the job has a substantial impact on the lives of others, inside or
outside the organisation.

• Autonomy: The degree of freedom, independence, and discretion the employee has in carrying out the work.

• Feedback: The degree to which the job provides clear, direct information about the effectiveness of
performance.

High levels of these characteristics increase the experienced meaningfulness, responsibility, and knowledge of
results, which in turn drive internal motivation, higher quality work, satisfaction, and lower absenteeism/turnover.

Decision Making, Leadership, and Organizational Structures

• Decision Making styles range from autocratic (manager decides alone) to consultative (manager seeks input)
to democratic/participative (team decides together) — the appropriate style depends on time pressure, the
team's expertise, and the importance of buy-in.

• Leadership styles include autocratic, democratic, and laissez-faire leadership; situational/contingency theories
argue effective leaders adapt their style to the maturity of the team and the nature of the task.

• Organizational structures: functional structure (grouped by discipline, e.g. testing, development),
project-based/pure project structure (dedicated project teams), and matrix structure (staff report to both a functional manager and a project manager) — each has trade-offs between specialisation, coordination, and speed of response.

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Part A: Software Effort Estimation.

1. Introduction to Software Effort Estimation

It is the process of predicting the effort, time, and cost required to develop or maintain a software system.

Estimation is done during project planning and revised as more information becomes available.

2. Problems with Overestimation an

2. Problems with Estimation

Underestimation

Causes schedule and cost overruns and puts pressure on the team.

May reduce software quality and damage client trust.

Overestimation

Causes resource and budget wastage.

May reduce competitiveness and lead to loss of contracts.

Key Point

The goal of estimation is to be accurate and realistic, not simply conservative or optimistic.

Basis of Software Estimation

Software size – LOC, Function Points, Object/Use-Case Points.

Historical data – previous similar projects.

Requirements and scope – clear requirements improve accuracy.

Resources and constraints – skills, tools, technology and deadlines.

Risk factors – complexity, new technology, external dependencies.

Development process – Waterfall, Agile, Incremental, etc.

Document all assumptions used for estimation.

4. Software Estimation Techniques

■ Expert Judgment and Estimation by Analogy use experience and comparison with previous projects.

■ Algorithmic/Parametric Models, Top-Down, Bottom-Up, and Price-to-Win are other commonly used techniques.

Algorithmic Models: Uses mathematical models such as COCOMO.

5. Expert Judgment & Delphi Technique

Expert Judgment uses the knowledge and experience of one or more experts to estimate effort, cost, and schedule.

Delphi Technique collects expert estimates anonymously in several rounds to reduce group bias and reach a consensus.

6. Estimation by Analogy

It compares a new project with similar completed projects and uses their actual effort as a starting point.

The historical effort is adjusted for differences in size, team experience, technology, or complexity.

Network Planning Models

A network planning model is a graphical technique used in project management to show the sequence, dependency, and duration of project activities.

1. PERT – Program Evaluation and Review Technique

PERT is used when activity times are uncertain and uses three time estimates: optimistic, most likely, and pessimistic.

It is useful for research, software development, and new projects where exact durations are difficult to predict.

Formula:

Te=To+4Tm+Tp6T_e=\frac{T_o+4T_m+T_p}{6}

2. CPM – Critical Path Method

CPM is used when activity durations are known or predictable and helps identify the critical path.

It calculates the minimum project duration and identifies activities having zero float/slack.

Example: A → B → C → D, where the longest-duration path is the critical path.

3. AON – Activity on Node

In AON, each node/box represents an activity, while arrows represent dependencies between activities.

It is also called the Activity-on-Node (AoN) or Precedence Diagramming Method (PDM).

Example:

[A: Design] → [B: Coding] → [C: Testing]

4. AOA – Activity on Arrow

In AOA, each arrow represents an activity, while nodes represent events or milestones.

Dummy activities may be required to represent logical dependencies correctly.

Example:

(1) ■■ A: Design ■■> (2) ■■ B: Coding ■■> (3)

5. Gantt Chart

A Gantt chart represents project activities as horizontal bars against a time scale, showing start time, finish time, and duration.

It is useful for project scheduling, progress tracking, and monitoring.

Example:

Activity Week 1 Week 2 Week 3 Week 4

Design ■■■■■

Coding ■■■■■■■■■■■

Testing ■■■■■

. Activity Planning

It is the process of identifying activities, their order, duration, and required resources.

It helps complete the project within the planned time and cost.

2. Project Schedule

It is a timetable showing the start and finish of each project activity.

It helps the project manager monitor progress and identify delays.

3. Steps in Project Scheduling

Identify activities, dependencies, resources, duration, and constraints.

Allocate resources and prepare the final schedule, usually using a Gantt chart.

4. Projects and Activities

A project is divided into smaller units called activities or tasks.

Activities consume time and resources and produce a specific result.

5. Work Breakdown Structure (WBS)

WBS divides a large project into smaller and manageable units of work.

It helps identify and organize all activities required for the project.

6. Characteristics of an Activity

An activity should have a clear start, end, duration, and outcome.

It normally requires resources such as people, equipment, time, or money.

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Risk Identification and Analysis

1. Risk Management

Risk management is the process of identifying, analysing, and controlling risks that may affect a project's cost, schedule, or quality.

It helps reduce the negative effects of unexpected events.

1. Risk Identification

Risk identification is the process of finding possible threats to a project before they occur.

Common techniques:

Checklists: Use a prepared list of common software project risks such as personnel, requirements, technology, and external dependencies.

Brainstorming: Team members discuss and list as many possible risks as possible, then organize and refine them.

3. Risk Analysis

Risk analysis determines the probability of a risk occurring and its impact/severity.

A probability–impact matrix is used to identify high-priority risks.

4. Risk Evaluation

Risk evaluation considers the combined effect of all risks on the project.

It compares the overall risk with the organisation's risk tolerance/threshold to decide whether the project is acceptable.

5. Monitoring and Control of Risk

Assign a risk owner to monitor each risk.

Set trigger points, regularly update the risk register, and apply suitable risk-reduction strategies.

Maintain contingency plans and budget/schedule buffers for unavoidable risks.

6. Decision Trees in Risk Evaluation

A decision tree shows decisions, possible events, probabilities, and outcomes.

Expected value is calculated as:

Expected Value = Probability × Outcome

It helps compare alternatives and select the option with the best expected value.

2. Risk Projection (Estimation)

Estimates the probability of each risk occurring.

Determines the impact/consequences of the risk on the project and product.

3. Risk Table Construction

A risk table lists each risk with its category, probability, and impact/severity.

High-probability and high-impact risks are given higher priority.

4. Risk Exposure (RE)

Risk exposure measures the expected loss from a risk.

Formula:

RE = Probability of Occurrence × Cost of Loss

Higher RE means the risk needs more attention and mitigation.

5. Risk Assessment and Refinement

Risk assessment: Compares risks with acceptable limits such as cost overrun, schedule delay, and performance loss.

Risk refinement: Breaks a broad risk into smaller, specific risks that are easier to manage and monitor.

4. Reducing the Risk

1. Hazard Prevention

Completely eliminate the risk or reduce it to a very low level.

Example: Schedule important staff meetings early to avoid staff unavailability.

2. Likelihood Reduction

Reduce the probability of a risk occurring through proper planning.

Example: Use prototypes to reduce late requirement changes.

3. Risk Avoidance

Change the project plan so that the project is no longer exposed to the risk.

Example: Reduce project scope to avoid missing a fixed deadline.

4. Risk Transfer

Transfer the impact of the risk to another party.

Example: Outsource a risky component or use insurance.

5. Contingency Planning

Prepare a backup/fallback plan if the risk occurs.

Example: Keep replacement programmers ready if key employees become unavailable.

Resource Allocation – Short Notes

1. Scheduling Resources

Assigns people, equipment, and tools to each project activity.

Checks resource availability and resolves conflicts between activities.

Uses Gantt charts/resource histograms to identify over-allocation or idle resources.

2. Critical Paths

Critical Path Method (CPM) identifies the sequence of activities that determines the minimum project duration.

Activities with zero slack/float form the critical path; any delay in them delays the whole project.

Uses Forward Pass (Earliest Start/Finish) and Backward Pass (Latest Start/Finish) to find the critical path.

3. Cost Scheduling

Spreads activity costs over the time periods when the costs will actually occur.

Helps manage cash flow, cost baseline, and planned vs. actual expenditure.

Supports time–cost trade-off, such as crashing critical activities to reduce project duration at extra cost.

Monitoring and Control – Short Notes

1. Creating a Framework for Monitoring and Control

Defines what to measure and how often, such as progress, milestones, cost, and quality.

Establishes reporting lines, tolerances, and change-control procedures.

Allows re-planning when actual project performance differs from the original plan.

2. Cost Monitoring

Compares actual expenditure with the planned cost regularly to detect problems early.

Uses Earned Value Analysis (EVA):

PV (Planned Value): Budgeted cost of scheduled work.

EV (Earned Value): Budgeted cost of completed work.

AC (Actual Cost): Actual cost spent on completed work.

Helps identify schedule and cost variances and provides early warning of budget overruns.

3. Prioritizing Monitoring

Monitoring is focused on activities that have the greatest impact on project success.

Critical-path activities: Closely monitored because delays affect the project completion date.

High-risk activities: Given more attention due to possible high impact.

High-cost activities: Tightly monitored to control budget overruns.

Near-milestone activities: Closely monitored as important deadlines approach.

. Categories of Software Project Risks

Project Risks: Affect project schedule and resources. Example: Staff turnover, unrealistic deadlines.

Product Risks: Affect software quality or performance. Example: Poor requirements, untested components.

Business Risks: Affect business viability or organizational success. Example: Product has no market demand, competitors release first.

Generic Risks: Common risks that can occur in most software projects. Example: Requirement changes, loss of key personnel.

Product-Specific Risks: Risks unique to the project's technology, architecture, or application domain.